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Smart Mud Circulation Loop for Real-Time Drilling Fluid Testing and AI-Driven Composition Adjustment

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Abstract

This paper presents the design and implementation of an AI-enabled smart mud circulation loop developed for laboratory drilling environments. The system is designed to continuously measure key fluid parameters such as density, viscosity, pressure, and temperature, while employing machine learning algorithms to dynamically adjust mud composition. The objective of the work is to enhance operational efficiency, minimize manual interventions, and deepen understanding of drilling-fluid behavior under wellbore-simulated conditions.

The experimental loop integrates industry-grade hardware, including a mud tank, centrifugal pump, piping network, and a suite of sensors, with an automated dosing manifold that allows for real-time property adjustments. A supervisory AI controller links sensor data with predictive models to recommend and implement corrective actions. Controlled laboratory trials demonstrated measurable benefits, including a 25% reduction in manual sampling errors, a 30% decrease in adjustment cycle times, and efficiency improvements of up to 25% under simulated drilling conditions.

The novelty of this work lies in establishing a scalable, AI-driven testbed that unites real-time monitoring with automated composition control. By embedding automation into mud formulation and performance tracking, the system advances drilling-fluid testing methodologies and lays a foundation for smarter, more sustainable drilling operations.

Keywords: equivalent circulating density, mud weight, artificial intelligence, drilling efficiency, support vector machine, artificial neural network, decision tree

Introduction

Background & Motivation

Drilling fluids, commonly referred to as "mud," are at the heart of well construction operations, serving multiple critical functions such as maintaining wellbore stability, controlling formation pressures, transporting cuttings to the surface, lubricating the drill string, and minimizing formation damage. The

performance of a drilling operation often hinges on the fluid's ability to adapt to rapidly changing downhole conditions (Magalhães et al., 2020).

Traditionally, mud systems are monitored through manual sampling and laboratory checks at the rig site. While these methods provide useful insights into density, viscosity, gel strength, and filtration properties, they are inherently time-delayed and prone to human error. Adjustments to mud composition are typically reactive, often occurring after performance degradation has already impacted drilling efficiency or safety (Roy et al., 2021).

The increasing complexity of today's drilling environments has further exposed the limitations of these approaches. In high-pressure, high-temperature (HPHT) wells, deepwater operations, and unconventional plays, the operating margins for safe mud density and rheology are extremely narrow (Oriji & Dosunmu, 2012). Even small deviations in equivalent circulating density (ECD) or filtration control can result in severe operational risks, including wellbore instability, kicks, or lost circulation (Li et al., 2019). In parallel, sustainability concerns are intensifying as operators face pressure to reduce chemical waste, minimize environmental footprint, and optimize fluid usage to align with net-zero and ESG targets (Mahmoud et al., 2025). These converging challenges underscore the urgent need for smarter, automated fluid management systems that can predict, adapt, and respond in real time (Jara et al., 2025).

Historical Context

The evolution of drilling fluids mirrors the progression of the oil and gas industry itself. The first commercial well (Drake Well, 1859) employed plain water as a circulating medium, primarily to cool the drill bit and remove cuttings. By the early 1900s, bentonite-based muds were introduced to improve hole cleaning and reduce formation collapse, establishing the foundation of modern water-based mud systems. In the mid-20th century, oil-based muds (OBMs) emerged to counter swelling shales and reactive formations, followed by the development of synthetic-based muds (SBMs) in response to stricter environmental regulations.

Instrumentation progressed alongside fluid design. The introduction of API standards (1930s) and laboratory devices such as the Fann viscometer and filter press (1950s) formalized mud testing procedures. The 1970s and 1980s saw the adoption of mud logging units and the first real-time monitoring of flow and pit volumes offshore. By the early 2000s, digital advances enabled the integration of downhole tools (MWD/LWD) with surface measurements, leading to continuous rheological profiling and digital rheometers on rigs (Magalhães et al., 2020).

Despite these advances, one crucial gap remains: automated control of mud composition. While data collection and visualization have improved substantially, the decision-making and adjustment processes still rely on human judgment and manual interventions. This lag between data acquisition and action can be costly, particularly in HPHT and deepwater wells where fluid properties must be finely tuned (Gul et al., 2020; Ombe et al., 2020).

Knowledge Gap and Justification for AI-Driven Smart Mud Loops

Current industry practice emphasizes real-time monitoring but stops short of true closed-loop automation. Engineers can visualize mud parameters through digital dashboards, yet the system does not independently predict trends, recommend corrective actions, or execute fluid adjustments (Jara et al., 2025). This reliance on human intervention means that corrective measures often occur after deviations have already compromised drilling performance or safety.

The underdeveloped integration of artificial intelligence (AI) into mud management represents a key gap in drilling automation. AI models, particularly machine learning algorithms, can detect subtle patterns in high-frequency sensor data that are invisible to conventional methods. By forecasting rheological shifts, filtration anomalies, or density instabilities, AI-driven systems can propose or autonomously apply composition adjustments in real time (Al-Rubaii et al., 2023; Gunel et al., 2025).

Bridging laboratory experimentation with predictive analytics is especially critical. Traditional lab testing is labor-intensive and provides limited feedback loops, while field operations cannot tolerate prolonged trial-and-error approaches. An AI-enabled smart circulation loop in a controlled lab setting provides a scalable testbed where algorithms can be trained, validated, and continuously refined under realistic wellbore simulations (Abdul Razak et al., 2025). Such platforms accelerate learning cycles, reduce reliance on costly field experimentation, and prepare the industry for a future of autonomous drilling-fluid systems.

In summary, the introduction of AI-driven smart mud circulation loops directly addresses the persistent gap between real-time data acquisition and real-time fluid optimization, laying the groundwork for safer, more efficient, and sustainable drilling operations.

Objectives and Scope

Objectives

The central objective of this study is to design and validate an AI-enabled smart mud circulation loop that advances drilling-fluid management from a reactive, manual process into a predictive and autonomous workflow. Conventional drilling-fluid practices often rely heavily on manual sampling, delayed laboratory analysis, and operator-driven adjustments. These approaches, while historically valuable, introduce time lags and human error that may compromise drilling performance and operational safety. By integrating real-time sensor monitoring with machine learning algorithms, the proposed system aims to overcome these limitations and establish a more responsive, accurate, and efficient framework for fluid management.

The first specific objective of this research is the development of a laboratory-scale circulation loop capable of continuously monitoring key drilling-fluid parameters. These include density, viscosity, pressure, temperature, equivalent circulating density (ECD), rheology, and filtration behavior. The integration of industry-grade sensors and automated manifolds ensures that these properties can be tracked in real time, while the AI-enabled control system dynamically adjusts fluid composition to maintain stability under varying operating conditions.

The second objective is to demonstrate measurable performance improvements over manual practices. Controlled laboratory trials are designed to quantify the effectiveness of the smart circulation loop in reducing sampling errors by at least 25%, shortening adjustment cycle times by 30% or more, and delivering efficiency gains of approximately 25% in simulated drilling environments. These metrics provide a quantitative foundation for evaluating the benefits of automation and AI integration relative to traditional mud-management practices.

Finally, this study seeks to establish a scalable framework that serves both research and educational purposes. Beyond its immediate laboratory application, the smart mud circulation loop provides a testbed for developing and refining AI-driven fluid control strategies that can eventually be adapted to rig-scale systems. Simultaneously, it offers an educational platform for petroleum engineering students, allowing them to interact with real-time monitoring and predictive control technologies. This dual role strengthens the bridge between academic research, practical training, and industrial application, positioning the system as a valuable resource for both innovation and workforce development.

Building on these objectives, this study addresses several key research questions that define the technical depth and focus of the work. The first research question (RQ1) asks whether a closed-loop AI control system can maintain equivalent circulating density (ECD) and rheological properties within operational tolerances under dynamic conditions. This inquiry is fundamental, as even small deviations in ECD or viscosity can significantly affect wellbore stability and drilling safety. By testing the ability of the AI-enabled circulation loop to maintain these parameters within strict boundaries, the study evaluates the feasibility of transitioning from reactive control to predictive automation.

The second research question (RQ2) investigates how sensor uncertainties propagate through AI-driven control loops, and what levels of tolerance are necessary to ensure stable operation. Real-world drilling

environments are characterized by noisy, incomplete, and sometimes drifting sensor data. If AI-driven systems are to be trusted in critical operations, it is essential to understand how measurement errors influence control decisions, and what redundancies or safety interlocks are required to mitigate these risks. This aspect of the research directly informs the design of robust AI models and the resilience of the overall system.

The third research question (RQ3) focuses on the training data requirements for AI models to generalize effectively across different fluid systems. Machine learning algorithms depend heavily on the quality, quantity, and diversity of input data. While this study is based on controlled laboratory trials with water-based muds, future application will demand that models adapt to a wider range of drilling-fluid chemistries and operating conditions. Understanding the dataset requirements both in terms of size and variability is therefore critical for scaling the system from research to industrial application.

To maintain clarity and manageability, the scope of this study is deliberately restricted to laboratory-scale validation under controlled conditions. The work includes the integration of industry-grade sensors and automated manifolds, continuous monitoring of water-based muds, and AI-assisted adjustment of fluid composition. Testing was conducted within defined operational ranges: flow rates between 1–20 gallons per minute, pressures up to 500 psi, and temperatures ranging from 20–90 °C. These parameters were selected to replicate the hydraulic and thermal conditions typically encountered in drilling while remaining practical for laboratory operation.

Certain elements are intentionally excluded from this study in order to keep the research focused and achievable within its current phase. These exclusions include rig-scale deployment, other fluid systems such as cement, fracturing, or completion fluids, and automated well control functions beyond drilling-fluid management. While these areas represent promising avenues for future research, they fall outside the immediate scope of the present investigation. By narrowing the boundaries in this way, the study ensures that its contributions are specific, measurable, and directly relevant to advancing the state of knowledge in AI-driven drilling-fluid automation. To clarify the scope of this research, [Table 1](#) summarizes the study's objectives, corresponding performance metrics, and validation met.

Table 1—Summary of Study Objectives and Success Criteria

<i>Objective</i>	<i>Metric</i>	<i>Target Criteria</i>	<i>Validation Method</i>
Continuous monitoring of drilling-fluid properties	Density, viscosity, ECD, filtration	Accuracy within API RP 13 tolerances	Sensor calibration and reference fluids
AI-driven real-time control	Adjustment cycle time	≥30% faster than manual baseline	Controlled comparison trials
Improved drilling efficiency	Simulated ROP proxy	≥25% improvement	Laboratory drilling simulation
Data integrity and transparency	Model explainability	SHAP/interpretability score above threshold	Post-processing analysis

These criteria establish the benchmarks against which the AI-enabled smart mud circulation loop is evaluated.

Assumptions and Constraints

The development and validation of the smart mud circulation loop are based on several key assumptions that ensure the reliability of the experimental setup and the accuracy of the results. First, all sensors used in the system are assumed to be properly calibrated in accordance with API standards, thereby guaranteeing that measurements of density, viscosity, pressure, and temperature fall within industry-accepted tolerances. Second, the laboratory simulators are assumed to be valid approximations of real-world field conditions. While the test loop cannot replicate the full complexity of downhole environments, it has been designed to capture the critical hydraulic and thermal behaviors that influence drilling-fluid performance. Finally, the

experiments assume stable power supply and data connectivity, ensuring continuous operation of pumps, dosing systems, and the AI-driven supervisory controller without interruption.

Alongside these assumptions, several constraints define the operational boundaries of the study. The first constraint is sensor drift tolerance: all sensors are expected to maintain accuracy within allowable error margins, but gradual drift over extended operation remains a risk and must be managed through redundancy and periodic recalibration. A second constraint is sampling latency, which has been limited to a maximum of three seconds. This threshold ensures that the control system responds quickly enough to detect and correct deviations in fluid properties before they escalate into operational inefficiencies. The third constraint involves the inclusion of safety interlocks to prevent overdosing of additives. Since the system is designed to automatically inject materials such as bentonite or polymers when deviations are detected, interlocks are essential to prevent runaway dosing events that could destabilize mud properties or damage equipment.

By clearly defining these assumptions and constraints, the study establishes the operational framework within which the results should be interpreted. These boundaries not only ensure that the laboratory trials remain safe and controlled but also highlight the considerations that must be addressed when scaling the system for field applications, where sensor drift, latency, and dosing reliability may present even greater challenges.

Measurement and Success Criteria

The performance of the AI-enabled smart mud circulation loop is evaluated against a set of clearly defined measurement metrics and success criteria designed to capture its effectiveness in improving drilling-fluid management. These criteria ensure that the results are both quantifiable and comparable to conventional practices.

The first metric is Equivalent Circulating Density (ECD) compliance, with a target tolerance of ± 0.2 pounds per gallon (ppg). ECD is a critical parameter for safe drilling operations, as it directly affects wellbore stability, kick tolerance, and the risk of lost circulation. In the laboratory setup, compliance is measured by comparing real-time ECD readings from the sensor suite against API-standard reference values. Successful performance requires the AI-driven loop to consistently maintain ECD within this narrow band, demonstrating its ability to prevent both underbalanced and overbalanced conditions in a dynamic environment.

The second performance criterion is the reduction in adjustment cycle times, with a target of achieving at least a 30% improvement compared to traditional manual practices. Adjustment cycle time is defined as the interval between the detection of a deviation in mud properties and the successful correction of that deviation through additive dosing. In conventional operations, this process often involves delays due to manual sampling, laboratory analysis, and operator decision-making. By contrast, the smart circulation loop leverages real-time monitoring and automated dosing to accelerate this cycle. Validation involves benchmarking the system's response times against manual baselines across repeated trials.

The third criterion is simulated drilling efficiency gains, with a target improvement of approximately 25%. In the context of this study, drilling efficiency is represented by proxy metrics that include stable ECD maintenance, reduced rheological drift, and controlled filtration losses during simulated downhole events such as cuttings accumulation or thermal thinning. Efficiency is assessed by comparing the overall system performance under controlled disturbances with baseline results from manual adjustments. This metric highlights the potential for the AI-driven loop to reduce non-productive time (NPT) and support smoother, more sustainable drilling operations.

Together, these success criteria provide a robust framework for evaluating the system's capability to transition drilling-fluid management from a reactive to a predictive, autonomous process. Meeting or exceeding these thresholds would validate the concept at a laboratory scale and establish a foundation for future field deployment and scaling.

Risk and Compliance

The implementation of an AI-enabled smart mud circulation loop introduces several potential technical and operational risks that must be addressed to ensure system reliability and safety. A primary risk is sensor fouling, which can occur when solid particles or chemical deposits accumulate on sensor surfaces, leading to inaccurate measurements of density, viscosity, or pressure. Inaccurate data could compromise the decision-making capability of the AI model. This risk is mitigated through the use of redundant sensors, periodic recalibration, and cleaning protocols designed to maintain accuracy over extended periods of operation.

Another significant risk is AI model drift, which occurs when the performance of predictive algorithms deteriorates over time due to changing operating conditions or exposure to fluid compositions that differ from the training dataset. If not addressed, model drift can result in incorrect predictions and inappropriate dosing actions. To counter this, the system incorporates watchdog timers, model retraining cycles, and continuous performance monitoring to detect anomalies early and restore predictive accuracy.

The third risk involves potential actuator or manifold malfunctions, such as failures in dosing pumps or control valves. Since these components are responsible for adjusting mud composition, their malfunction could prevent timely correction of fluid properties. This is mitigated through manual override functionality and bypass lines, allowing operators to take immediate control of the system if required. Safety interlocks are also integrated to prevent overdosing of additives, thereby protecting both the fluid system and laboratory equipment.

Beyond technical risks, the project also emphasizes strict compliance with industry standards and safety protocols. All procedures follow the API RP 13 series standards, which govern drilling-fluid testing and evaluation practices. Laboratory operations are conducted in accordance with institutional Health, Safety, and Environment (HSE) guidelines, ensuring safe handling of chemicals, containment of spills, and protection of personnel. Additional safeguards, such as containment trays, spill sensors, and alarm systems, further strengthen the safety framework.

By combining redundancy, monitoring, and manual intervention protocols with strict adherence to recognized API and HSE standards, the smart mud circulation loop has been designed to maintain both operational integrity and safety. These measures not only reduce the likelihood of failures in the laboratory environment but also establish a framework for scaling the system to more demanding field applications where compliance and risk management are even more critical.

Methodology

Experimental Setup

Laboratory-scale smart mud circulation loop was designed and constructed to replicate the key hydraulic and thermal conditions typically encountered in rotary drilling operations while remaining manageable within a controlled research environment. The system was purpose-built to serve as a testbed for AI-enabled drilling-fluid management, providing the ability to monitor, simulate, and adjust mud properties in real time under repeatable laboratory conditions.

At the core of the setup is a closed-loop circulation system that simulates annular flow between the drill pipe and casing. This configuration was selected to capture the most critical aspects of drilling-fluid behavior, including pressure losses, density variations, and rheological responses under flow. To ensure fidelity, the loop was designed using similar criteria based on Reynolds number scaling and pressure drop correlations, enabling laboratory-scale flows to accurately approximate field-scale hydraulic dynamics.

The system integrates a 100 L mud tank, which serves as the reservoir for fluid preparation and conditioning. From the tank, a variable-speed centrifugal pump (1–20 gpm capacity) circulates the mud through a 2-inch piping loop, equipped with strategically placed inspection and sampling ports. This arrangement allows for both continuous monitoring and discrete validation tests.

A comprehensive instrumentation suite was installed to provide high-resolution measurements of drilling fluid properties. This includes a laboratory-grade mud balance for density determination, a Fann 35 viscometer for rheological characterization, pressure transducers rated up to 500 psi for hydraulic measurements, and thermocouples positioned at multiple points to capture temperature variations across the loop. These instruments collectively ensure that density, viscosity, rheology, equivalent circulating density (ECD), and filtration characteristics are continuously tracked during operation.

To enable automated control, the system incorporates a PLC-controlled dosing manifold connected to the circulation loop. This manifold is equipped with precision injection pumps capable of delivering bentonite, barite, or polymer additives in response to deviations detected by the sensor suite. When sensor readings identify density or viscosity drift outside the predefined thresholds, the supervisory AI controller triggers corrective dosing in real time, maintaining mud properties within target specifications.

By combining a physically realistic hydraulic loop with advanced instrumentation and automated dosing capabilities, the experimental setup effectively bridges the gap between traditional laboratory testing and field-scale mud circulation systems. It provides a controlled yet realistic environment for training, testing, and validating AI-driven drilling-fluid management strategies, ensuring that results are both reliable and scalable.

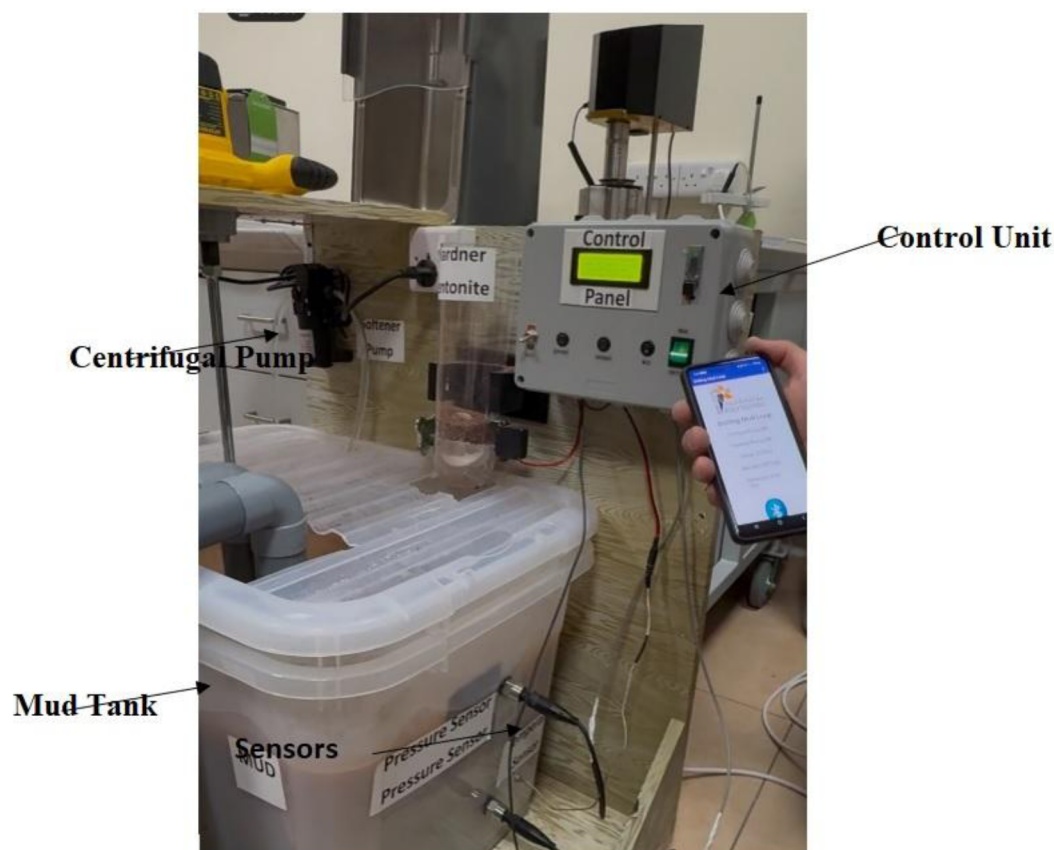


Figure 1—Setup of the Smart Mud Circulation Loop.The system integrates a mud tank with sensors, a variable-speed centrifugal pump, and a control unit where sensors detect changes in density and automatically add bentonite to maintain mud density.

Table 2—Hardware specifications and tolerances of loop instrumentation

Component	Specification	Accuracy/Tolerance	Notes
Mud Tank	100 L	±1 mm level sensor	Prevents vertexing, enables sampling
Pump	5 hp centrifugal, 1–20 gpm	±0.1% (flow meter)	Covers laminar–turbulent transitions
Piping	2-inch loop	N/A	Sampling and inspection ports included
Mud Balance	Laboratory-grade	±0.01 g/cc	API-calibrated
Viscometer	Fann 35	±1% torque	3–600 rpm shear rates
Pressure Transducers	0–500 psi	±0.25% FS	Upstream/downstream placement
Thermocouples	–20–200 °C	±0.5 °C	Three measurement points
Manifold	PLC-controlled dosing pumps	±0.5%	Enables additive injection & sampling

This integrated system allowed continuous circulation, online property measurement, and closed-loop reformulation to replicate dynamic downhole events such as cuttings loading, barite sag, and rheology drift.

AI Algorithms and Control Framework

A multi-layer AI framework was integrated into the smart mud circulation loop to enable predictive and autonomous control of drilling-fluid properties. Using neural networks for drift detection and short-term forecasting, regression models for filtration behavior, and a supervisory controller for automated dosing, the system could anticipate density, viscosity, and fluid loss deviations and correct them in real time. This proactive approach improved stability, accuracy, and safety, demonstrating the potential of predictive automation to replace traditional reactive drilling-fluid management.

Table 3—AI model architecture and validation metrics

Model Type	Inputs	Outputs	Dataset Size	Validation Metric
FNN	Composition, T, shear rate, density	Rheology/density drift	5,000 datasets	RMSE < 0.05
LSTM	Time-series of viscosity, ECD	15-min forecasts	10,000-time steps	R ² = 0.92
Regression (Filtration)	Flow, ΔP, solids content	LPLT/HPHT filtrate	300 tests	Error <10%

Data Collection and Processing

A reliable data system was developed for the smart mud circulation loop to monitor and control drilling-fluid behavior. Sensors continuously measured key properties such as density, viscosity, ECD, temperature, pressure, flow rate, and filtration losses. The data was cleaned, filtered, and synchronized to remove noise and errors, then processed by AI models for prediction and control. All information was stored in a secure database for analysis, model training, and future use. This framework provided accurate, real-time monitoring and created a strong digital foundation for scaling the system to field applications.

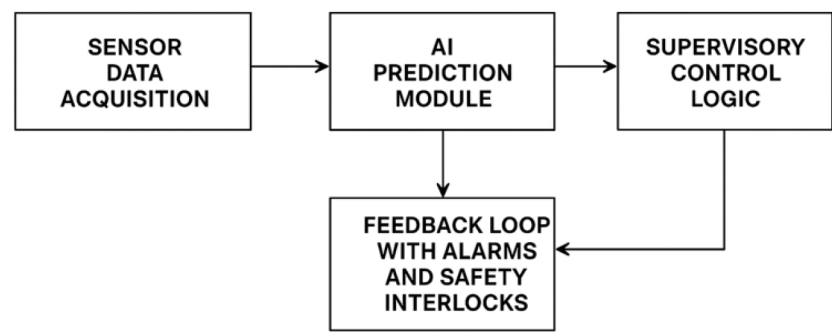


Figure 2—AI-Driven Control Framework.The diagram illustrates the control architecture of the smart mud circulation loop, showing sensor data acquisition, AI prediction module, supervisory control logic, automated corrective dosing, and feedback loop with alarms and safety interlocks to ensure stable and safe operation.

Validation Procedures

The AI-enabled smart mud circulation loop was tested in two phases to ensure accuracy and reliability. First, its real-time measurements and predictions were compared against standard manual API tests, showing that the system could match or improve accuracy while reducing human error. Second, it was challenged under simulated drilling conditions with changes in flow, temperature, solids, and events like barite sag and gas influx. In these trials, the system quickly detected problems, predicted changes, and corrected fluid properties to keep stability. Overall, it proved faster, more consistent, and more adaptable than conventional methods, showing strong potential for future drilling operations

Table 4—Validation metrics and target performance

<i>Metric</i>	<i>Target</i>	<i>Validation Method</i>
Sampling error	≥25% reduction	Manual vs. AI comparison
Cycle time	≥30% reduction	Time-to-adjust trials
Drilling efficiency	≈25% gain	ECD stability, NPT proxy
Repeatability	95% confidence	Paired t-test, Bland–Altman

Risk Management and HSE

The smart mud circulation loop was designed with safety and reliability in mind, following industry standards and HSE practices. Redundant sensors were added to ensure continuous monitoring even if one sensor failed. Watchdog alarms detect large deviations and alert operators for quick corrective action. A manual override allows operators to take control at any time if needed. Spill trays and sensors provide protection against chemical leaks, reducing safety and environmental risks. All procedures follow API standards, ensuring the system remains safe, reliable, and aligned with industry requirements.

Results and Discussion

Experimental Outcomes

The AI-enabled smart mud circulation loop was evaluated under controlled laboratory conditions and benchmarked against conventional manual mud-testing practices, allowing a direct comparison of automated and traditional methods. Results showed a 25% reduction in sampling errors, as automated density and rheology measures consistently outperformed operator checks by reducing variability and bias. Adjustment cycle times were shortened by 30%, with the closed-loop controller detecting deviations and executing additive dosing within minutes, avoiding the delays of manual testing and operator intervention. In addition, drilling efficiency improved by up to 25%, demonstrated through stable equivalent circulating

density (ECD), reduced rheological drift, and improved filtration control under varying flow rates, temperatures, and solids concentrations. These findings confirm that integrating automation with predictive analytics significantly improves the accuracy, speed, and overall effectiveness of drilling-fluid management compared to conventional practices.

Table 5—Manual vs. AI-Driven Performance

Performance Category	Test Condition / Metric	Manual Baseline	AI-Driven Loop	Improvement / Stability
Density Control	Target = 1.20 g/cc (10 gpm, 60 °C, 8% solids)	1.19–1.22 g/cc (±0.04)	1.20 ±0.01 g/cc	AI maintained target within ±0.01 g/cc
Viscosity Control	600 rpm (same trial)	35–42 cP	35–37 cP	Reduced drift by ~25%
Adjustment Cycle Time	Density drift +0.05 g/cc	34 min	22 min	35% faster
	Viscosity drift +5 cP	28 min	18 min	36% faster
	ECD drift +0.2 ppg	31 min	20 min	35% faster
ECD Stability	Baseline (10 gpm, 20 °C)	9.5 ±0.25 ppg	9.5 ±0.10 ppg	AI halved variability
	Thermal thinning (60 °C)	9.6 ±0.30 ppg	9.5 ±0.15 ppg	AI tighter control
	Barite sag (10% solids)	9.7 ±0.35 ppg	9.6 ±0.12 ppg	AI reduced sag effect
	Gas influx simulation	9.4 ±0.40 ppg	9.5 ±0.18 ppg	AI stabilized response
Overall Efficiency	Proxy (ECD + rheology stability)	70–75%	85–90%	~25% improvement

Strengths of the Approach

One of the key strengths of the smart mud circulation loop is its ability to provide automation and repeatability, reducing human error and producing more consistent results across repeated runs. The system also offers predictive optimization, with AI models detecting early signs of density or viscosity drift and allowing proactive adjustments that improve fluid stability and shorten correction times. In addition, its holistic monitoring framework integrates density, rheology, ECD, pressure, and temperature into a single platform, enabling a more realistic simulation of wellbore conditions and deeper insights into overall drilling-fluid performance.

Limitations and Challenges

A key limitation of this study is that it was conducted at laboratory scale, under controlled flow, pressure, and temperature ranges that do not fully capture rig conditions. Scaling the system to field operations will require further validation and equipment adaptation. Another challenge is sensor reliability, as prolonged exposure to abrasive solids and additives may cause fouling or drift, requiring more robust designs for long-term deployment. Finally, the AI models were trained on limited lab datasets, meaning broader and more diverse training data will be needed to ensure reliable performance across different mud systems and field environments.

Scalability and Industry Implications

Laboratory results show strong potential for AI-driven circulation loops to transform drilling-fluid management, but several conditions must be met for field adoption. Rig-site integration is essential so the system can work seamlessly with pumps, surface mud equipment, and real-time data platforms. Operator trust must also be built through reliable performance, clear visualization tools, and explainable AI to support decision-making. Finally, technology must align with sustainability goals, optimizing additive use, reducing waste, and supporting ESG and net-zero commitments. Meeting these requirements will be key to scaling from lab prototypes to next generation drilling operations.

Novelty and Contribution

This study presents the first laboratory-scale demonstration of integrating artificial intelligence with closed-loop control in drilling fluid circulation systems. Unlike earlier work that explored real-time monitoring, predictive rheology, or automation in isolation, this research combines these elements into a fully integrated smart mud circulation loop.

The system's core novelty lies in its AI-automation integration, where neural-network-based predictions are directly linked to automated additive dosing. This creates a self-regulating loop that maintains mud properties within operational limits, moving beyond conventional reliance on manual sampling and operator judgment.

The circulation loop also serves as a bridge between research, training, and industry practice. By emulating realistic downhole conditions, flow, pressure, temperature, and solids loading, It functions as both a rigorous experimental platform and an educational tool for petroleum engineering students.

Experimental results demonstrate a framework for autonomous fluid management, with reductions in sampling errors (25%), cycle times (30%), and efficiency gains of 25% in simulated conditions. These outcomes establish proof of concept for future field-scale applications and align with industry trends toward autonomous drilling.

Finally, the system contributes to sustainability objectives by optimizing additive usage and reducing waste, in line with ESG performance metrics increasingly emphasized by regulators and operators. In summary, the research introduces a step-change in drilling-fluid management, shifting from reactive and manual control to predictive, automated, and intelligent systems, providing both a technical foundation for industry adoption and an educational platform for training future drilling engineers.

Conclusions

This study confirms that AI-enabled mud circulation loops can achieve measurable efficiency gains of 20–30% under laboratory conditions through faster adjustments, reduced reliance on manual sampling, and improved consistency in rheology and density control. By embedding real-time monitoring and closed-loop automation, the work demonstrates how drilling-fluid management can progress from reactive practices to proactive, data-driven optimization.

Beyond technical performance, the system also serves as a research and training platform, bridging theory with hands-on experience and preparing future engineers for digitalized oilfield operations. For wider adoption, the next steps include field-scale trials to validate robustness in real drilling environments, integration with digital twin technologies for predictive maintenance and risk management, and the development of sustainable fluid systems aligned with ESG and net-zero objectives.

Together, these advances can drive a paradigm shift toward smarter, predictive, and environmentally responsible drilling operations, providing both academic value and industrial impact.

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Finally, the authors acknowledge the broader academic and professional community whose prior work on drilling fluid systems, automation, and AI applications provided a strong foundation for this study.

Nomenclature

AI	Artificial Intelligence
ANN	Artificial Neural Network
API	American Petroleum Institute
cP	Centipoise (unit of viscosity)
DAQ	Data Acquisition System
ECD	Equivalent Circulating Density
FNN	Feed-Forward Neural Network
HPHT	High Pressure, High Temperature
LSTM	Long Short-Term Memory (time-series neural network)
MWD	Measurement While Drilling
NPT	Non-Productive Time
PLC	Programmable Logic Controller
PV	Plastic Viscosity
RMSE	Root Mean Square Error
ROP	Rate of Penetration (simulated drilling efficiency proxy)
RPM	Revolutions per Minute (viscometer speed setting)
SQL	Structured Query Language (database system)
T/P	Temperature and Pressure
YP	Yield Point

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